Responsible & Safe AI Systems — 2025 Assignments

Questions and answers, week-wise. Answers are as marked in the NPTEL answer videos; questions marked (Select all that apply) have more than one correct option.

Week 0

Q1. What is the definition of Artificial Intelligence?

a) a programming language used in data analysis
b) a type of hardware designed for computing
c) a software used to generate text/media
d) the ability for a computer to imitate human behavior

Answer: d) the ability for a computer to imitate human behavior

Q2. Which type of machine learning uses a combination of a small amount of labeled data and a large amount of unlabeled data to improve learning accuracy?

a) Reinforcement Learning
b) Unsupervised Learning
c) Supervised Learning
d) Semi-Supervised Learning

Answer: d) Semi-Supervised Learning

Q3. Which of the following is true? Select all that apply. **(Select all that apply)**

a) A model with high bias leads to underfitting of the model.
b) A model with high variance leads to overfitting of the model.
c) High variance and high bias lead to a more optimal model.
d) Decreasing model complexity results in low bias but high variance.

Answer: a) A model with high bias leads to underfitting of the model.; b) A model with high variance leads to overfitting of the model.

Q4. What is an example of a biased AI system? Select all that apply.

a) An AI-powered hiring algorithm that ranks resumes based on ethnicity of the candidate.
b) A chatbot that fails to understand slang or technical jargon.
c) A recommendation system that shows different ads to different users based on search history.
d) A weather prediction system that fails to predict hurricanes due to insufficient data.

Answer: a) An AI-powered hiring algorithm that ranks resumes based on ethnicity of the candidate.

Q5. What does "transparency" mean in an AI system?

a) Making the AI unknown to users.
b) Explaining how the AI made the decision in a clear way.
c) Using very little data as input to the AI system.
d) Optimizing the AI system for speed.

Answer: b) Explaining how the AI made the decision in a clear way.

Q6. What is the goal of responsible AI?

a) To make AI faster.
b) To reduce computation.
c) To make AI adaptable.
d) To ensure AI behaves in a fair and ethical way.

Answer: d) To ensure AI behaves in a fair and ethical way.

Q7. What is one risk when using an AI system “unethically”?

a) The AI system may become too slow.
b) The AI system may become human.
c) The AI system causes harm.
d) None of the above.

Answer: c) The AI system causes harm.

Q8. What is the purpose of a validation dataset?

a) To train the model on new data.
b) To test the final accuracy of the model.
c) To tune the hyperparameters and prevent overfitting.
d) All of the above.

Answer: c) To tune the hyperparameters and prevent overfitting.

Q9. What is the bias-variance tradeoff in machine learning?

a) A balance between the training dataset size and testing dataset size.
b) Trying to increase bias and variance to improve model accuracy.
c) Finding a balance between simplicity and complexity of a model resulting in optimal performance.
d) Deciding between supervised and unsupervised learning.

Answer: c) Finding a balance between simplicity and complexity of a model resulting in optimal performance.

Q10. Which of the following is an example of supervised learning?

a) An AI learns to play a game through trial and error.
b) A classification model trained on images labelled “cats” and “dogs”.
c) An AI that groups customers together based on their purchases.
d) A chatbot that remembers past conversations.

Answer: b) A classification model trained on images labelled “cats” and “dogs”.

Q11. What is the definition of “algorithmic bias”?

a) When an AI system makes decisions faster than humans.
b) When a model prioritizes bias over variance.
c) When an AI system produces unfair or discriminatory outcomes due to biased data or design
d) When the model only runs on certain devices.

Answer: c) When an AI system produces unfair or discriminatory outcomes due to biased data or design

Q12. What is the difference between supervised and unsupervised learning?

a) Supervised learning uses labeled data, while unsupervised learning does not.
b) Supervised learning is used only for robotics, while unsupervised learning is for software.
c) Supervised learning requires no data, while unsupervised learning requires large datasets.
d) Supervised learning is always faster than unsupervised learning.

Answer: a) Supervised learning uses labeled data, while unsupervised learning does not.

Week 1

Q1. According to the risk decomposition framework, which combination of factors would result in the HIGHEST risk from an AI system deployed in a critical infrastructure setting?

a) Low vulnerability, high hazard exposure, low hazard severity
b) High vulnerability, low hazard exposure, high hazard severity
c) High vulnerability, high hazard exposure, high hazard severity
d) Low vulnerability, low hazard exposure, high hazard severity

Answer: c) High vulnerability, high hazard exposure, high hazard severity

Q2. The concept of treacherous turns in AI systems refers to:

a) AI systems making computational errors during complex calculations
b) AI systems behaving differently once they reach sufficient intelligence
c) AI systems being hacked by malicious actors
d) AI systems consuming too much computational power

Answer: b) AI systems behaving differently once they reach sufficient intelligence

Q3. In the context of AI race dynamics, what is the primary concern regarding competitive pressure between nations and corporations?

a) It will make AI systems too expensive for general use
b) It will result in compatible AI standards globally
c) It will slow down AI innovation and progress
d) It may lead to rushed development that compromises safety measures

Answer: d) It may lead to rushed development that compromises safety measures

Q4. The "Swiss cheese model" mentioned in organizational risks suggests that:

a) Organizations should have a single, very strong safety measure
b) Safety measures should be implemented randomly across the organization
c) Multiple layers of defense compensate for individual weaknesses
d) Safety measures are unnecessary if the AI system is well-designed

Answer: c) Multiple layers of defense compensate for individual weaknesses

Q5. Which scenario best illustrates the concept of proxy gaming?

a) An AI chess program that cheats by accessing opponent's strategy
b) A recommendation system optimizing for user engagement rather than user well-being
c) An AI translator that produces grammatically incorrect sentences
d) A facial recognition system that fails to identify certain ethnic groups

Answer: b) A recommendation system optimizing for user engagement rather than user well-being

Q6. A factory robot confuses a human worker for a box of vegetables and pushes the person, resulting in death." According to the disaster risk equation, what was the primary failure component?

a) Hazard (misclassification capability)
b) Hazard Exposure (human-robot proximity)
c) Vulnerability (employee safety protocols)
d) All components failed equally

Answer: a) Hazard (misclassification capability)

Q7. According to the risk taxonomy presented, malicious use of AI differs from rogue AI primarily in that:

a) Malicious use involves intentional harmful deployment by humans, while rogue AI acts independently
b) Malicious use only affects cybersecurity, while rogue AI affects all domains
c) Malicious use is easier to detect than rogue AI behavior
d) Malicious use requires more advanced AI capabilities than rogue AI

Answer: a) Malicious use involves intentional harmful deployment by humans, while rogue AI acts independently

Q8. Deceptive Alignment in AI systems is:

a) AI systems that are openly hostile to humans
b) AI systems that appear to be following instructions but are actually pursuing different goals
c) AI systems that cannot understand human language properly
d) AI systems that work too slowly to be effective

Answer: b) AI systems that appear to be following instructions but are actually pursuing different goals

Q9. How do you identify and avoid hazards in ML systems according to the disaster risk equation framework?

a) Alignment
b) Robustness
c) Monitoring
d) Systemic Safety

Answer: b) Robustness

Q10. Red teaming in AI safety primarily serves to:

a) Accelerate model training
b) Identify system vulnerabilities
c) Improve computational efficiency
d) Reduce inference latency

Answer: b) Identify system vulnerabilities

Q11. Which technique is most effective for detecting deceptive alignment?

a) Training the model with more than 1000 samples
b) Mechanistic interpretability
c) Increasing model parameters
d) Reward modeling

Answer: b) Mechanistic interpretability

Q12. RoBERTa succeeds in reasoning tasks where BERT fails due to:

a) Better tokenization
b) Emergent capabilities from scaling
c) Improved attention mechanisms
d) Larger vocabulary size

Answer: b) Emergent capabilities from scaling

Week 2

Q1. Which of the following is true?

a) Extremistan has thin tails while Mediocristan has long tails

b) Mediocristan distributions are harder to predict than Extremistan

c) In Extremistan, the total is determined by a few large events with tyranny of the accidental

d) Extremistan has mild randomness while Mediocristan has wild randomness

Answer: c) In Extremistan, the total is determined by a few large events with tyranny of the accidental

Q2. What is the key difference between covariate shift and concept shift in distribution shifts?

a) Covariate shift changes P(y|x) while concept shift changes P(x)

b) Covariate shift changes P(x) while P(y|x) remains constant, concept shift changes P(y|x) while P(x) remains constant

c) Both change P(x) and P(y|x) simultaneously

d) Covariate shift affects labels while concept shift affects features

Answer: b) Covariate shift changes P(x) while P(y|x) remains constant, concept shift changes P(y|x) while P(x) remains constant

Q3. In the AugMix methodology, what is the primary advantage over uncontrolled random augmentations?

a) It uses skip connections to keep images recognizable while applying diverse augmentations

b) It requires less computational power

c) It only applies single augmentations instead of multiple

d) It focuses on geometric transformations only

Answer: a) It uses skip connections to keep images recognizable while applying diverse augmentations

Q4. In the context of adversarial attacks, what does "transferability" specifically refer to?

a) The ability to transfer attacks from one domain to another

b) The ability to transfer defenses across different architectures

c) The ability to convert white-box attacks to black-box attacks

d) The ability of adversarial examples crafted for one model to work on other models

Answer: a) The ability to transfer attacks from one domain to another

Q5. Black Swan lies in which of the following categories?

a) Known Knowns

b) Known Unknowns

c) Unknown Knowns

d) Unknown Unknowns

Answer: d) Unknown Unknowns

Q6. Which of the following are valid approaches for defending against adversarial attacks? (Select all that apply) **(Select all that apply)**

a) Data augmentation techniques
b) Adversarial training using adversarial examples during training
c) Using more data and larger models
d) Reducing model complexity to avoid overfitting
e) Adversarial pretraining on larger datasets like ImageNet

Answer: a) Data augmentation techniques; b) Adversarial training using adversarial examples during training; e) Adversarial pretraining on larger datasets like ImageNet

Q7. In the RLHF optimization objective, why is a KL-divergence penalty term added to the reward maximization?

a) To prevent the model from generating repetitive outputs
b) To ensure the model stays close to the original pretrained model
c) To improve the computational efficiency of the training process
d) To increase the diversity of generated samples

Answer: b) To ensure the model stays close to the original pretrained model

Q8. What does "reward hacking" specifically refer to in the context of RLHF?

a) Humans providing incorrect feedback to manipulate the system
b) External attackers compromising the reward model
c) The reward model overfitting to the training data
d) The model finding ways to maximize the reward function without achieving the intended behavior

Answer: d) The model finding ways to maximize the reward function without achieving the intended behavior

Q9. Identify the equations that can lead to a long-tailed distribution.

a) Idea * student * resources * time
b) Idea * student + resources * time
c) Idea + student + resource + time
d) Idea - student * resource - time

Answer: a) Idea * student * resources * time

Q10. What is the primary advantage of using pairwise comparisons over direct scalar ratings in human feedback collection?

a) Pairwise comparisons are faster to collect
b) They require fewer human annotators
c) Human judgments are noisy and miscalibrated, but pairwise comparisons are more reliable
d) They provide more granular feedback information

Answer: c) Human judgments are noisy and miscalibrated, but pairwise comparisons are more reliable

Q11. What is a major challenge with using a single reward function in RLHF?

a) It is computationally expensive to optimize
b) It cannot represent a diverse society of humans
c) It requires too much training data
d) It is unstable during training

Answer: b) It cannot represent a diverse society of humans

Q12. What is Direct Preference Optimization (DPO) equivalent to in the context of RLHF component removal?

a) RLHF - Human Feedback

b) RLHF - Reward Model

c) RLHF - RL

d) RLHF - Policy Optimization

Answer: b) RLHF - Reward Model

Week 3

Q1. What is the definition of "Machine Unlearning"?

a) Removing the influences of training data from a trained model
b) The ability of a machine learning model to adapt to new data
c) A technique used to compress machine learning models
d) The ability of a machine learning model to learn a variety of data

Answer: a) Removing the influences of training data from a trained model

Q2. What might be some types of information that one might want to remove from model data? (Select all that apply.) **(Select all that apply)**

a) Private data
b) Toxic or unsafe content
c) Accurate information
d) Model hyperparameter settings
e) Stale knowledge

Answer: a) Private data; b) Toxic or unsafe content; e) Stale knowledge

Q3. What is GDPR's Article 17 about in the context of Machine Learning?

a) Right to be remembered
b) Right to be modified
c) Right to be distributed
d) Right to be forgotten

Answer: d) Right to be forgotten

Q4. What are the steps in the SISA approach?

a) Sampled, Isolated, Stopped, Aggregated
b) Sharded, Imitate, Sliced, Annotated
c) Sharded, Isolated, Sliced, Aggregated
d) Sampled, Imitate, Stopped, Annotated

Answer: c) Sharded, Isolated, Sliced, Aggregated

Q5. What is Membership Inference Attack (MIA) used for?

a) To train LLMs faster
b) To improve the models robustness
c) To remove accurate data
d) To classify between training and unseen data

Answer: d) To classify between training and unseen data

Q6. What is the role of differential privacy?

a) To improve model accuracy
b) To make the model forget everything
c) To make models indistinguishable with/without certain data points
d) To speed up training time

Answer: c) To make models indistinguishable with/without certain data points

Q7. Which benchmarks are used to evaluate unlearning in LLMs? (Select all that apply.) **(Select all that apply)**

a) TOFU
b) GLUE
c) WMDP
d) GUIDE

Answer: a) TOFU; c) WMDP; d) GUIDE

Q8. Case: A hospital wants to share patient health data with research institutions to support public health studies. However, it must ensure that a patient's identity cannot be inferred. To achieve this, the hospital generates and sends aggregate statistics to […] institute. Which form of unlearning is this?

a) Exact unlearning
b) Unlearning via differential privacy
c) Just ask for unlearning
d) Empirical Unlearning

Answer: b) Unlearning via differential privacy

Q9. Which isn't a type of graph unlearning?

a) Node unlearning
b) Edge unlearning
c) Label unlearning
d) Node feature unlearning

Answer: c) Label unlearning

Q10. Which methods are used for Node unlearning? (Select all that apply.) **(Select all that apply)**

a) GraphEraser
b) GUIDE
c) Projector
d) GraphdEditor
e) GNNDelete
f) MEGU

Answer: a) GraphEraser; e) GNNDelete; f) MEGU

Q11. What are “hidden representations”?

a) The final model outputs
b) Raw training data stored in memory
c) Intermediate activation vectors captured during model execution
d) The model's loss values during training

Answer: c) Intermediate activation vectors captured during model execution

Week 4

Q1. As per the lecture, in the context of Machine Learning, what is the definition of 'bias'?

a) Systematic deviation from rationality and judgement
b) Systematic error in the collection, analysis, or interpretation of data
c) Systematic favoritism or discrimination towards certain groups/outcomes
d) Systematic behaviour when solving complex tasks

Answer: c) Systematic favoritism or discrimination towards certain groups/outcomes

Q2. Why did Microsoft's Tay chatbot become offensive shortly after it was launched?

a) It learned toxic behaviour from user interactions on Twitter.
b) It was hacked by a rival company.
c) It was trained on outdated information.
d) It was programmed to be controversial for publicity.

Answer: a) It learned toxic behaviour from user interactions on Twitter.

Q3. As per the lecture, which of the following is a/are a category of bias? (Select all that apply.) **(Select all that apply)**

a) Gender
b) Race
c) Scientific Facts
d) Profession
e) Currency exchange rates

Answer: a) Gender; b) Race; d) Profession

Q4. According to the ML Pipeline, what may be a source of bias? (Select all that apply.) **(Select all that apply)**

a) Annotators beliefs
b) Hardware used for computation
c) Balanced dataset
d) Biased training data

Answer: a) Annotators beliefs; d) Biased training data

Q5. What does the Legal Safety Score (LSSβ) represent?

a) The model's ability to predict legal outcomes based solely on accuracy.
b) A metric combining fairness and accuracy using a β-weighted harmonic mean.
c) A score based on the usage of legal jargon for marginalized groups.
d) An evaluation metric used to define how well a model understands legal jargon.

Answer: b) A metric combining fairness and accuracy using a β-weighted harmonic mean.

Q6. What do LLMs use to prevent harmful outputs?

a) Data augmentation
b) Guardrails
c) Faster GPUs
d) Dropout layers

Answer: b) Guardrails

Q7. Which of the following is true about bias?

a) It never exists.
b) It sometimes exists.
c) It only exists in American-created models.
d) It always exists.

Answer: d) It always exists.

Q8. A researcher is evaluating a facial recognition model they helped develop. During testing, they select images where the model performs better, such as images with ideal lighting or frontal faces, while ignoring diverse or difficult cases (like low-light, non-white faces, or side angles). Which type of bias is this?

a) Reporting bias
b) Sampling bias
c) Experimenter's bias
d) Historical bias

Answer: c) Experimenter's bias

Q9. What is the issue with evaluating models only based on accuracy?

a) Accuracy only reflects hardware performance.
b) Accuracy doesn't reveal bias.
c) Accuracy checks for fairness.
d) Accuracy changes the labelling.

Answer: b) Accuracy doesn't reveal bias.

Q10. Why might using Western-aligned datasets be problematic for the Indian demographic?

a) The datasets are too large which may slow down training time.
b) They are written in a different language.
c) They don't reflect Indian social/societal norms.
d) They contain too many low-resolution images.

Answer: c) They don't reflect Indian social/societal norms.

Q11. What is a 'stereotype'?

a) A factual statement that applies to all humans.
b) A scientifically proven characteristic.
c) A legal rule used to govern a society.
d) A widely held belief about some group/entity.

Answer: d) A widely held belief about some group/entity.

Q12. What does the CrowS-Pairs dataset contain?

a) Pairs of sentences that differ only in minimally distant social bias.
b) Dialogues between humans and a chatbot.
c) Pairs of biased and unbiased images.
d) Code snippets with and without bugs.

Answer: a) Pairs of sentences that differ only in minimally distant social bias.

Q13. What is sampling bias?

a) When historical data reflects inequalities that existed in the world at that time.

b) When data is collected from a completely random and diverse group.

c) If proper randomization is not used during data collection.

d) When a model builder keeps training a model until it produces a result that aligns with their original hypothesis.

Answer: c) If proper randomization is not used during data collection.

Week 5

Q1. What is considered an ideal Stereotype Score (ss)?

a) 0%
b) 25%
c) 50%
d) 75%

Answer: a) 0%

Q2. What does SEAT stand for?

a) Standard Evaluation Assessment Test
b) Semantic Evaluation Annotation Test
c) Structured Embedding Accuracy Test
d) Sentence Embedding Association Test

Answer: d) Sentence Embedding Association Test

Q3. In counterfactual data augmentation (CDA), what is altered to rebalance the corpus?

a) Sentence Length
b) Syntax
c) Bias attribute words
d) Vocabulary complexity

Answer: c) Bias attribute words

Q4. Which characteristic makes a language model more likely to generate gender-neutral responses in text-to-text tasks? (Select all that apply.) **(Select all that apply)**

a) Training on diverse and balanced datasets.
b) Use of bias-specific adapter modules.
c) Conditioning outputs on explicit gender tokens.
d) Reliance on pretrained token embeddings without fine-tuning.

Answer: a) Training on diverse and balanced datasets.; b) Use of bias-specific adapter modules.

Q5. Which toolkit is used to add programmable guardrails to LLM-based conversational applications like ChatGPT?

a) GPT-4
b) NVIDIA NeMo
c) CoDi
d) MAFIA

Answer: b) NVIDIA NeMo

Q6. Which of the following is the correct formula for Pointwise Mutual Information (PMI)?

a) PMI(wi, wj) = log2 [N · c(wi, wj) / (c(wi)^2 · c(wj)^2)]
b) PMI(wi, wj) = log2 [c(wi) · c(wj) / (N · c(wi, wj))]
c) PMI(wi, wj) = log2 [N · c(wi, wj) / (c(wi) · c(wj))]
d) PMI(wi, wj) = log2 [c(wi, wj) / (N · c(wi) · c(wj))]

Answer: c) PMI(wi, wj) = log2 [N · c(wi, wj) / (c(wi) · c(wj))]

Q7. 'Useful fairness' couples which of the following? (Select all that apply.) **(Select all that apply)**

a) Context awareness
b) Bias Score
c) STS task performance
d) Dataset diversity

Answer: a) Context awareness; c) STS task performance

Q8. What is the key architectural idea behind the MAFIA model for effective debiasing?

a) Fusing bias-specific adapters while keeping the base model the same.
b) Replacing all model weights with debiased adapters.
c) Dynamically routing inputs based on detected bias type.
d) Using GANs to hallucinate fair outputs.

Answer: a) Fusing bias-specific adapters while keeping the base model the same.

Q9. Which of the following is true about proprietary models?

a) They are more neutral compared to CoDi and other open source models.
b) They are less neutral compared to CoDi and other open source models.
c) They have more bias than CoDi and other open source models.
d) They have the same neutrality as CoDi and other open source models.

Answer: a) They are more neutral compared to CoDi and other open source models.

Q10. Which of the following are benchmark datasets commonly used to measure bias in language models? (Select all that apply.) **(Select all that apply)**

a) Stereoset
b) CrowS-Pairs
c) ImageNet
d) Bias-STS-S
e) SQuAD

Answer: a) Stereoset; b) CrowS-Pairs; d) Bias-STS-S

Q11. What is 'gender-bleaching' in the context of VLMs?

a) Improving the quality of input images.
b) Turning all people in the input images white.
c) Enhancing gender-specific features in input images.
d) Removing/Obscuring visual cues related to gender in input images.

Answer: d) Removing/Obscuring visual cues related to gender in input images.

Q12. Why might a single adapter for all bias types (like iDEBall) fail?

a) Cannot effectively debias across all categories.
b) Trains slower than other models.
c) Requires more input data.
d) Does not understand contextual information.

Answer: a) Cannot effectively debias across all categories.

Week 6

Q1. What are the primary disadvantages of cryptographic solutions for privacy protection? Select all that apply. **(Select all that apply)**

a) Inference possibility remains uncertain in all scenarios
b) Utility reduces
c) It is expensive
d) Security guarantees are absolute in all cases

Answer: b) Utility reduces; c) It is expensive

Q2. Why do anonymization techniques often fail to provide adequate privacy protection?

a) They require excessive computational resources
b) Adversaries can leverage auxiliary databases for de-anonymization attacks
c) They only work with structured data formats
d) The anonymization process introduces too much noise

Answer: b) Adversaries can leverage auxiliary databases for de-anonymization attacks

Q3. In the randomized response model, what does the parameter epsilon in e^ε represent?

a) The percentage of truthfulness of the response that is allowed to be revealed
b) The ratio between falsehood and randomness in responses
c) The computational complexity of the algorithm
d) The number of participants in the study

Answer: a) The percentage of truthfulness of the response that is allowed to be revealed

Q4. What characterizes an ideal privacy model?

a) Maximum utility with zero privacy guarantees
b) Perfect privacy with minimal utility
c) Balanced trade-off between utility and privacy requirements
d) Complete data suppression for absolute privacy

Answer: c) Balanced trade-off between utility and privacy requirements

Q5. Given: Xi represents truth, Yi represents randomized value of Xi, and Zi = (Yi − (1/(1+e^ε))) × (e^ε+1)/(e^ε−1). What is the expected value Zi?

a) E[Zi] = Yi
b) E[Zi] = Xi
c) E[Zi] = e^ε × Xi
d) E[Zi] = 0

Answer: b) E[Zi] = Xi

Q6. In the randomized response model, which statements are correct? Select all that apply. **(Select all that apply)**

a) Privacy guarantee can be controlled by parameter ε
b) Privacy and utility are independent of sample size
c) Higher ε values provide stronger utility
d) Utility guarantee scales with √n

Answer: a) Privacy guarantee can be controlled by parameter ε; c) Higher ε values provide stronger utility; d) Utility guarantee scales with √n

Q7. Consider datasets: X = {x1, x2, ..., xN} (truth), Y = {y1, y2, ..., yN} (revealed values). To derive better estimators Z = {z1, z2, ..., zN} from Y, What process is required?

a) Removing bias from Y introduced through randomization
b) Adding bias to Y removed through randomization
c) Removing variance from Y introduced through randomization
d) Adding variance to Y removed through randomization

Answer: a) Removing bias from Y introduced through randomization

Q8. Which factors directly influence the privacy guarantee in differential privacy mechanisms? Select all that apply. **(Select all that apply)**

a) Sensitivity of the query function
b) Magnitude of added noise
c) Size of the dataset
d) Privacy parameter Epsilon

Answer: a) Sensitivity of the query function; b) Magnitude of added noise; d) Privacy parameter Epsilon

Q9. Which trust scenario correctly describes the differential privacy model?

a) Trust the curator; Trust the world
b) Do not trust the curator; Trust the world
c) Trust the curator; Do not trust the world
d) Do not trust the curator; Do not trust the world

Answer: b) Do not trust the curator; Trust the world

Q10. For fixed privacy levels, how do sample size requirements differ between Laplacian mechanism and Randomized response?

a) Constant factor difference
b) Exponential factor difference
c) Logarithmic factor difference
d) Quadratic factor difference

Answer: a) Constant factor difference

Q11. Higher privacy guarantees can be achieved in which scenarios? Select all correct options. **(Select all that apply)**

a) When Epsilon parameter is increased
b) When noise magnitude is increased
c) When inverse sensitivity is high
d) When variance in the mechanism is high
e) When utility requirements are maximized

Answer: b) When noise magnitude is increased; d) When variance in the mechanism is high

Q12. In the context of randomized response, what happens to utility as privacy guarantees increase?

a) Utility remains constant
b) Utility increases proportionally
c) Utility decreases due to increased noise
d) Utility becomes undefined

Answer: c) Utility decreases due to increased noise

Week 7

Q1. In approximate differential privacy, what role does the delta (δ) parameter play? (Notation same as in the lecture)

a) It ensures that epsilon differential privacy holds for all possible sets
b) It allows for some rare events (set S) where epsilon differential privacy may not hold
c) It reduces the amount of noise required in all cases
d) It eliminates the need for the epsilon parameter

Answer: b) It allows for some rare events (set S) where epsilon differential privacy may not hold

Q2. What is a key advantage of approximate differential privacy over standard differential privacy?

a) It provides stronger privacy guarantees
b) It eliminates the need for noise addition
c) It can increase utility
d) It works only with categorical data

Answer: c) It can increase utility

Q3. In approximate differential privacy, the Gaussian noise follows which pattern?

a) $N(0, \sigma)$ where $\sigma = \sqrt{(d \log(1/\delta))/(n+\varepsilon)}$
b) $N(0, \sigma)$ where $\sigma = \sqrt{(d \log(1/\delta))/(\varepsilon)}$
c) $N(0, \sigma)$ where $\sigma = d \log(1/\delta)/(n-\varepsilon)$
d) $N(0, \sigma)$ where $\sigma = \sqrt{(d \log(1/\delta))/(n \cdot \varepsilon)}$

Answer: b) $N(0, \sigma)$ where $\sigma = \sqrt{(d \log(1/\delta))/(\varepsilon)}$

Q4. What does Statistical Parity require for a fair classifier?

a) The classifier should have equal accuracy across all groups
b) The probability of positive predictions should be equal across protected and non-protected groups
c) The true positive rates should be identical for all demographic groups
d) Individual similar cases should receive similar predictions

Answer: b) The probability of positive predictions should be equal across protected and non-protected groups

Q5. For an ideal fair algorithm under Equality of Opportunity, what condition must be satisfied?

a) The overall prediction rates must be equal across groups
b) The false positive rates of unprivileged and privileged groups should be equal
c) The true positive rates of unprivileged and privileged groups should be equal
d) The precision should be identical for both protected and non-protected groups

Answer: c) The true positive rates of unprivileged and privileged groups should be equal

Q6. What is the Post-Processing property in differential privacy?

a) Any preprocessing of data before applying DP mechanisms maintains the privacy guarantee
b) Privacy guarantees are strengthened when multiple post-processing steps are applied
c) Any data-independent transformation applied to the output of a differentially private mechanism does not degrade its privacy guarantee
d) Post-processing can be used to improve the privacy guarantee of any mechanism

Answer: c) Any data-independent transformation applied to the output of a differentially private mechanism does not degrade its privacy guarantee

Q7. In a PCA analysis, if the reconstruction error for female data points is lower than for male data points, what does this indicate?

a) The dataset is biased against females
b) The dataset is biased against males
c) Reconstruction error differences are due to random noise
d) Male data is more correlated

Answer: d) Male data is more correlated

Q8. In the exponential mechanism to calculate the price to maximize the revenue, identify the correct statement in the scenario where 2 unequal prices result in the same revenue:

a) Both prices have an unequal probability of being selected
b) Both prices have an equal probability of being selected
c) A higher price has a higher probability of being chosen due to normalisation
d) A lower price has a higher probability of being chosen due to normalisation

Answer: b) Both prices have an equal probability of being selected

Q9. In an ideal situation where the models are completely fair, the different parity values are:

a) Approach 0
b) 1
c) Approach 1
d) 0

Answer: a) Approach 0

Q10. In a classifier, if a data point lies exactly on the decision boundary (hyperplane), what is the probability of it belonging to the positive class?

a) Greater than 50%
b) Less than 50%
c) Equal to 50%
d) Cannot be determined from the given information

Answer: c) Equal to 50%

Q11. In Fair Logistic Regression, the equation P(M(x)=1|C=1) - P(M(x)=1|C=0) represents which fairness metric?

a) Equality of Opportunity
b) Statistical Parity
c) Predictive Parity
d) Individual Fairness

Answer: b) Statistical Parity

Week 8

Q1. In pixel-attribution methods, which statement best distinguishes perturbation-based approaches from gradient-based saliency?

a) Perturbation-based methods compute input gradients; gradient-based methods fit local surrogate models.

b) Perturbation-based methods are model-agnostic and computationally expensive; gradient-based methods use model internals and are faster.

c) Perturbation-based methods always produce sparser explanations; gradient-based methods always produce denser explanations.

d) Perturbation-based methods require access to intermediate activations; gradient-based methods are strictly black-box.

Answer: b) Perturbation-based methods are model-agnostic and computationally expensive; gradient-based methods use model internals and are faster.

Q2. Which behavior of a saliency map under layer-randomization is a red flag that the map is not capturing learned model structure?

a) The saliency map remains largely unchanged after randomizing many layers.

b) The saliency map changes dramatically as earlier layers are randomized.

c) The saliency map's sign (positive/negative) flips while spatial structure remains.

d) Absolute saliency values shrink while layout shifts.

Answer: a) The saliency map remains largely unchanged after randomizing many layers.

Q3. Which of the following is fully equivariant to translation and rotation:

a) StyleGAN2

b) ProtoPNet

c) StyleGAN3

d) None of the above

Answer: c) StyleGAN3

Q4. Optimized-mask saliency methods (optimize a mask that when applied alters prediction) are brittle because:

a) Their masks always converge to a single-pixel trigger.

b) They require closed-form solutions for mask gradients.

c) Results strongly depend on mask initialization and hyperparameters.

d) They are invariant to adversarial perturbations.

Answer: c) Results strongly depend on mask initialization and hyperparameters.

Q5. For token-level gradient saliency in text, a large gradient magnitude for a token most reliably indicates:

a) The token is causally required for the model's prediction.

b) The model's output is sensitive to small embedding perturbations at that token.

c) The token is semantically important to humans.

d) The token was the only one used during training for that class.

Answer: b) The model's output is sensitive to small embedding perturbations at that token.

Q6. Which of the following is NOT a name commonly associated with pixel attribution methods?

a) Saliency map
b) Sensitivity map
c) Feature attribution
d) Convolution map

Answer: d) Convolution map

Q7. Which of the following are realistic attack vectors for implanting Trojans into neural networks? (Select all that apply)

a) Poisoning a fraction of a public training dataset with triggers and target labels.
b) Fine-tuning a model on carefully curated clean data only.
c) Distributing pretrained models via model libraries that already contain hidden functionality.
d) Applying purely random weight perturbation after training without trigger examples.
e) Injecting triggers only at inference time without altering training data or model weights.

Answer: a) Poisoning a fraction of a public training dataset with triggers and target labels.; c) Distributing pretrained models via model libraries that already contain hidden functionality.

Q8. Defenses that reverse-engineer potential triggers (e.g., via optimization) rely on which observations? (Select all that apply)

a) One can optimize small masks/patterns that cause misclassification to a target label.
b) If an optimized trigger for a particular label is significantly smaller/simpler than others, it suggests a Trojan.
c) Reverse-engineered triggers always exactly match the original poisoning trigger used at training.
d) Pruning neurons highly activated by a reverse-engineered trigger can mitigate the Trojan.
e) Training a meta-classifier to detect Trojaned models is computationally cheap and always generalizes.

Answer: a) One can optimize small masks/patterns that cause misclassification to a target label.; b) If an optimized trigger for a particular label is significantly smaller/simpler than others, it suggests a Trojan.; e) Training a meta-classifier to detect Trojaned models is computationally cheap and always generalizes.

Q9. How does Guided BackProp differ from standard backpropagation in generating saliency maps?

a) It only considers positive gradients by zeroing out negative activations and gradients.
b) It back propagates gradients with all activations zeroed out.
c) It focuses on highlighting both negative and positive contributions.
d) It requires padding 1 to the image before backpropagation.

Answer: a) It only considers positive gradients by zeroing out negative activations and gradients.

Q10. In the context of mechanistic interpretability, what do inhibitory connections in neural circuits primarily accomplish?

a) They amplify signal strength between neurons
b) They create redundant pathways for information flow
c) They reduce the probability of information transfer between neurons
d) They store long-term memory patterns

Answer: c) They reduce the probability of information transfer between neurons

Q11. What is the primary limitation of LIME's local explanations?

a) LIME only works on image data

b) LIME explanations are locally faithful but not necessarily globally consistent

c) LIME requires access to model internals

d) LIME cannot handle categorical features

Answer: b) LIME explanations are locally faithful but not necessarily globally consistent

Week 9

Q1. Which of the following is NOT mentioned as one of the RAI (Responsible AI) principles in the slides?

a) Fairness
b) Explainability
c) Sustainability
d) Privacy

Answer: c) Sustainability

Q2. According to the lecture, how many Executive Orders on AI were issued?

a) One
b) Two
c) Three
d) Four

Answer: b) Two

Q3. What does PEMAT stand for in the context of healthcare video assessment?

a) Patient Education Materials Assessment Tool
b) Patient Evaluation Medical Assessment Test
c) Public Education Medical Assessment Tool
d) Patient Educational Material Analysis Tool

Answer: a) Patient Education Materials Assessment Tool

Q4. According to the presentation, which domains need AI emergency response capabilities?

a) Only .com domains
b) Only .gov domains
c) .mil, .com and .gov domains
d) Only .edu domains

Answer: c) .mil, .com and .gov domains

Q5. Which of the following statement(s) is true regarding the development and implementation of AI systems?

a) Policy considerations and technical details are equally important
b) Technical details alone are sufficient for effective AI development
c) Policy considerations are only important in few countries
d) AI systems do not require any policy or ethical considerations

Answer: a) Policy considerations and technical details are equally important

Q6. In the suffix attacks, what is the vulnerability related to?

a) Data poisoning
b) Model stealing
c) Breaking alignment policies of chatbots
d) Privacy breaches

Answer: c) Breaking alignment policies of chatbots

Q7. What type of multi-disciplinary approaches are required for AI evaluation?

a) Only computer science approaches
b) Only engineering approaches
c) Social science, engineering, and computer science approaches
d) Only social science approaches

Answer: c) Social science, engineering, and computer science approaches

Q8. Based on the lecture content: From a mathematical perspective, which of the following is not considered a major problem in AI today, compared among others?

a) Explainability
b) Hallucinations
c) Data privacy
d) Bias

Answer: b) Hallucinations

Q9. According to the lecture, who is primarily responsible for self-regulation in the context of AI and technology?

a) Individuals / Individual organizations
b) Government agencies
c) Only the organizations with more than 1 crore turnover
d) International organizations

Answer: a) Individuals / Individual organizations

Q10. What is the six-step process for adopting a systems approach to fairness?

a) Define, Measure, Understand, Improve, Mitigate, Monitor
b) Design, Build, Test, Deploy, Evaluate, Maintain
c) Plan, Execute, Review, Adjust, Scale, Optimize
d) Collect, Process, Analyze, Model, Validate, Implement

Answer: a) Define, Measure, Understand, Improve, Mitigate, Monitor

Q11. In the Accuracy vs. Disparity chart, what represents the Ideal model?

a) High accuracy, high disparity
b) Low accuracy, low disparity
c) High accuracy, low disparity
d) Moderate accuracy, moderate disparity

Answer: c) High accuracy, low disparity

Week 10

Q1. What does ‘AGI’ stand for?

a) Augmented General Intelligence
b) Artificial Guided Intelligence
c) Augmented Guided Intelligence
d) Artificial General Intelligence

Answer: d) Artificial General Intelligence

Q2. According to the lecture, why is ‘blackbox access’ insufficient for AI agents?

a) It is too expensive to implement for most companies.
b) It poses a significant security risk for the company that developed the AI.
c) It prevents auditors from understanding the model's internal workings and decision-making processes.
d) It doesn't allow for the assessment of the model's training data.

Answer: c) It prevents auditors from understanding the model's internal workings and decision-making processes.

Q3. Which of the following best describes one of the possible definitions of AGI?

a) AI systems that are limited to specific tasks.
b) AI systems surpassing human intelligence.
c) AI models trained for basic automation.
d) AI systems which show high training accuracy.

Answer: b) AI systems surpassing human intelligence.

Q4. What is the key difference between 'reward gaming' and 'goal miss generalization' in AI systems?

a) Reward gaming is an intentional exploitation of the reward function, while goal miss generalization is an accidental misinterpretation of the goal.
b) Reward gaming is easier to detect and correct than goal miss generalization.
c) Reward gaming involves optimizing the wrong reward function, while goal miss generalization involves optimizing a correlated but incorrect reward function.
d) Reward gaming is a more significant threat to AI safety than goal miss generalization.

Answer: a) Reward gaming is an intentional exploitation of the reward function, while goal miss generalization is an accidental misinterpretation of the goal.

Q5. According to the lecture, what is the primary concern regarding the arrival of AGI?

a) AGI will be primarily used for malicious purposes by bad actors and rogue states.
b) AGI will be too expensive to develop and maintain, leading to a new form of global inequality.
c) The arrival of AGI could lead to chaotic power struggles, the extinction of humanity, or other severe negative consequences.
d) AGI will lead to massive job displacement and economic disruption.

Answer: c) The arrival of AGI could lead to chaotic power struggles, the extinction of humanity, or other severe negative consequences.

Q6. As mentioned in the lecture, what is the main concern about the use of social media data for training AI models?

a) The data is often of low quality, which leads to inaccurate and unreliable AI models.

b) There is a lack of incentive to protect user privacy in the current business models of tech companies.

c) The data is not diverse enough, which leads to biased and unfair AI models.

d) The data is too expensive for smaller companies to acquire, leading to a monopolization of AI development.

Answer: d) The data is too expensive for smaller companies to acquire, leading to a monopolization of AI development.

Q7. According to the lecture, what 'dangerous amount of power' do AI assistants create for the companies that develop them?

a) The power to censor information and control the user's access to knowledge.

b) The power to manipulate users' beliefs and behaviors based on their personal information and habits.

c) The power to control the user's personal finances and investments.

d) The power to replace human workers in a wide range of industries.

Answer: d) The power to replace human workers in a wide range of industries.

Q8. Which of the following are identified as significant challenges or concerns related to the development and deployment of advanced AI? (Select all that apply.) **(Select all that apply)**

a) The difficulty of achieving global coordination to pause or regulate AI development, similar to challenges like climate change.

b) The potential for AI systems to be misused by individuals for personal entertainment, leading to decreased productivity.

c) The legal and ethical issues surrounding the use of copyrighted material for training Large Language Models (LLMs).

d) The risk of "goal miss generalization," where an AI optimizes for a correlated but incorrect goal, leading to unintended and potentially harmful behavior.

e) The high computational cost of training advanced AI models, making them accessible only to a few large tech companies.

Answer: a) The difficulty of achieving global coordination to pause or regulate AI development, similar to challenges like climate change.; c) The legal and ethical issues surrounding the use of copyrighted material for training Large Language Models (LLMs).; d) The risk of "goal miss generalization," where an AI optimizes for a correlated but incorrect goal, leading to unintended and potentially harmful behavior.; e) The high computational cost of training advanced AI models, making them accessible only to a few large tech companies.

Q9. According to the lecture, what is the main legal challenge when an AI system makes a biased mistake in an employment context?

a) It is difficult to prove that the AI was the sole cause of the mistake.

b) Current laws primarily hold the employer responsible, not the AI developer.

c) AI developers are protected by international treaties.

d) There are no existing laws that cover discrimination in hiring.

Answer: b) Current laws primarily hold the employer responsible, not the AI developer.

Q10. What is meant by 'implicit bias' in an AI system?

a) Unconscious and involuntary stereotypes and assumptions embedded within algorithms.

b) Biases that are intentionally programmed into the AI by developers.

c) Biases that only appear when the AI interacts with specific users.

d) A safety feature that helps the AI avoid making biased statements.

Answer: a) Unconscious and involuntary stereotypes and assumptions embedded within algorithms.

Q11. Which of the following are presented as key areas of concern or focus for the future of AI? (Select all that apply) **(Select all that apply)**

a) The need for international regulation based on fundamental human rights.

b) The difficulty in marketing AI products to a skeptical public.

c) The environmental impact, particularly energy and water consumption.

d) The challenge of assigning legal responsibility when AI systems cause harm.

Answer: a) The need for international regulation based on fundamental human rights.; c) The environmental impact, particularly energy and water consumption.; d) The challenge of assigning legal responsibility when AI systems cause harm.

Q12. What is the primary danger of the 'arms race' mentality in AI development?

a) It slows down innovation by focusing too much on safety.

b) It encourages collaboration and open-sourcing of models.

c) It leads to rapid, unchecked development without adequate time for regulation and safety checks.

d) It reduces the profitability of AI companies.

Answer: c) It leads to rapid, unchecked development without adequate time for regulation and safety checks.

Q13. How is 'multimodality' viewed in the context of achieving AGI?

a) It has already been fully achieved by current AI models.

b) It is seen as essential for an AI to match the full range of human cognitive abilities.

c) It is considered a distraction from the more efficient text-based training methods.

d) It is believed to be the primary cause of bias in AI systems.

Answer: b) It is seen as essential for an AI to match the full range of human cognitive abilities.

Week 11

Q1. What are the countermeasures for computer security? (Select all that apply) **(Select all that apply)**

a) Train the users

b) Wait until the threat goes away on its own.

c) Silently eliminate the threat.

d) Warn users about the threat.

Answer: a) Train the users; d) Warn users about the threat.

Q2. Which of the following is an OECD AI Principle? (Select all that apply.) **(Select all that apply)**

a) International co-operation for trustworthy AI

b) Transparency and Explainability

c) Deletion of AI systems

d) Removing guardrails from AI

e) Robustness. Security and Safety

f) Investing in AI research and development

Answer: a) International co-operation for trustworthy AI; b) Transparency and Explainability; e) Robustness. Security and Safety; f) Investing in AI research and development

Q3. According to the AI Value Chain, who uses AI systems under authority?

a) Provider

b) Deployer

c) Distributor

d) Representative

Answer: b) Deployer

Q4. According to the OECD AI principles, what is 'Accountability'?

a) AI actors should respect the rule of law, human rights, democratic and human-centered values throughout the AI system lifecycle.

b) AI actors should commit to transparency and responsible disclosure regarding AI systems.

c) AI systems should be robust, secure, and safe throughout their entire lifecycle so that, in conditions of normal use, foreseeable use or misuse, or other adverse conditions, they function appropriately and do not pose unreasonable safety and/or security risks.

d) AI actors should be accountable for the proper functioning of AI systems and for the respect of the above principles, based on their roles, the context, and consistent with the state of the art.

Answer: d) AI actors should be accountable for the proper functioning of AI systems and for the respect of the above principles, based on their roles, the context, and consistent with the state of the art.

Q5. According to the EU AI Act, what AI Systems are considered "Unacceptable Risk"?

a) AI systems that understand too many languages.

b) AI systems that can understand logic but not emotional statements.

c) AI systems that can generate media such as images, videos, and audio.

d) AI systems that are able to behaviorally manipulate people.

Answer: d) AI systems that are able to behaviorally manipulate people.

Q6. According to Semantic Graph Entropy, which of the following is true?

a) Entropy and consistency are not related.
b) Entropy and consistency are proportional.
c) Entropy and consistency are inversely proportional.
d) Consistency is constant regardless of entropy.

Answer: c) Entropy and consistency are inversely proportional.

Q7. What is the main problem identified with Large Language Models (LLMs)?

a) They are too slow.
b) They are inconsistent.
c) They lack creativity.
d) They cannot identify human emotions.

Answer: b) They are inconsistent.

Q8. What does “Semantic Consistency” mean?

a) The ability to understand different languages.
b) The ability to make consistent decisions.
c) The ability to generate grammatically correct sentences.
d) The ability to learn new vocabulary.

Answer: b) The ability to make consistent decisions.

Q9. According to the lecture, what type of scenarios do LLMs "struggle" with more?

a) Commonsense scenarios.
b) Mathematical problems.
c) Moral scenarios.
d) Factual questions.

Answer: a) Commonsense scenarios.

Q10. What was a key finding of the SaGE framework regarding consistency and accuracy?

a) Consistency and accuracy are the same problem.
b) Consistency and accuracy are not the same problem and need separate evaluation.
c) Accuracy automatically implies consistency.
d) Consistency is irrelevant if accuracy is high.

Answer: b) Consistency and accuracy are not the same problem and need separate evaluation.

Q11. What is "Rule of Thumb" generation?

a) Developing general guidelines from practical experience.
b) A method for generating paraphrases.
c) A general guideline for model training.
d) A statistical measure of model performance.

Answer: a) Developing general guidelines from practical experience.

Week 12

Q1. According to the lecture, what is the primary limitation of graphs as data structures?

a) They can only model high-risk applications.
b) They are difficult to interpret.
c) They can only model pairwise relationships between nodes.
d) They cannot be used for fraud detection.

Answer: c) They can only model pairwise relationships between nodes.

Q2. What is a “p-cell” in a cell complex?

a) An element of dimension p.
b) A vertex (0-dimensional).
c) An edge (1-dimensional).
d) A general graph.

Answer: a) An element of dimension p.

Q3. How does a cell complex overcome the limitation of graphs?

a) By using more complex algorithms.
b) By capturing interactions between multiple nodes.
c) By reducing computational costs.
d) By simplifying the graph structure.

Answer: b) By capturing interactions between multiple nodes.

Q4. What does FORGE stand for?

a) Framework For Real-time Graph Explanations.
b) Fundamental Operations for Relevant Graph Embeddings.
c) Framework For Higher-Order Representations In Graph Explanations.
d) Fast Optimization of Graph Explanations.

Answer: c) Framework For Higher-Order Representations In Graph Explanations.

Q5. Based on the "Lifting the Graph" algorithm, which of the following are represented as augmented nodes? (Select all that apply.) **(Select all that apply)**

a) All nodes (0-cells)
b) Edges (1-cells)
c) Cycles (2-cells)
d) Boundary relations

Answer: a) All nodes (0-cells); b) Edges (1-cells); c) Cycles (2-cells)

Q6. What do Language Models (LMs) primarily train to predict?

a) The previous token.
b) The next token.
c) Grammatical structure.
d) Semantic relationships.

Answer: b) The next token.

Q7. According to the lecture, what is a “guarded” attribute in the context of representations?

a) An attribute that is easily classified.
b) An attribute that is protected from manipulation.
c) An attribute that cannot be classified based on the representations.
d) An attribute that enhances model performance.

Answer: c) An attribute that cannot be classified based on the representations.

Q8. What is the goal of “Affine Concept Erasure”?

a) To enhance a particular attribute.
b) To remove all attributes from a representation.
c) To apply a transformation that guards a particular attribute.
d) To make representations more easily interpretable.

Answer: c) To apply a transformation that guards a particular attribute.

Q9. When making vectors from one distribution to look like those of another (e.g., toxic to non-toxic), what is a key desired outcome besides guarding?

a) Increasing the model's complexity.
b) Preserving semantics unrelated to the changed attribute.
c) Introducing new semantic meanings.
d) Reducing the dimensionality of the vectors.

Answer: b) Preserving semantics unrelated to the changed attribute.

Q10. What is a “Black Swan”?

a) A common, predictable event.
b) An unforeseen event with extreme consequences.
c) An event that happens frequently in training data.
d) An event with minor consequences.

Answer: b) An unforeseen event with extreme consequences.

Q11. What is "Probing"?

a) A method to physically modify neural networks.
b) A method to analyze information stored in a model's representations.
c) A technique for speeding up model training.
d) A way to generate new data for models.

Answer: b) A method to analyze information stored in a model's representations.

Q12. What is the term for hidden functionality implanted into models by adversaries that can cause dangerous changes in behavior when triggered?

a) Trojan
b) Worm
c) Swarm
d) Backdoor

Answer: d) Backdoor